DSA REFERENCE

Trie Failure Modes

Common wrong turns, debugging cues, and complexity traps.

Store strings by shared prefixes so lookup cost depends on key length rather than number of stored keys.

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DSA / TRIE

DSA

Trie Failure Modes

CORE IDEA

Store strings by shared prefixes so lookup cost depends on key length rather than number of stored keys.

Recognition cues

+ prefix queries
+ dictionary word search
+ autocomplete
+ many string lookups with shared prefixes

Implementation sequence

+ choose child representation
+ mark terminal nodes
+ separate exact and prefix search
+ consider memory tradeoffs

Invariant

The path from root to a node spells one prefix; an end marker distinguishes full words from prefixes.

Worked reasoning

Autocomplete follows the query prefix, then enumerates terminal descendants subject to a result limit.

Complexity

O(L) insert and lookup for key length L; memory is proportional to stored prefix nodes.

Failure modes

- [ ] forgetting end markers
- [ ] treating prefix as a full word
- [ ] oversized fixed alphabets
- [ ] unbounded enumeration after prefix match

Practice ladder

Trie Failure Modes: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Implement Trie**	State the cue, invariant, edge case, and proof before implementation.
2	**Design Add and Search Words**	State the cue, invariant, edge case, and proof before implementation.
3	**Word Search II**	State the cue, invariant, edge case, and proof before implementation.
4	**Replace Words**	State the cue, invariant, edge case, and proof before implementation.
5	**Search Suggestions System**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.